

Building Better Guidelines: QA Problems and Remediation in Russian-Language Audio Annotation

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Introduction

Dataset quality depends not only on annotation accuracy, but on how faithfully the dataset reflects the real-world language environment the model is being trained to handle. A quality control system poorly adapted to the nature of the task can systematically impoverish a dataset — not through annotator error, but through the pressure of the evaluation system itself.

This document describes recurring problems in QA processes for Russian-language audio annotation projects: underspecified instructions that create an illusion of clarity; subjective criteria applied without calibration; taxonomic failures in classification systems. Each of these problems individually reduces inter-annotator agreement and degrades evaluation quality. Together, they trigger a mechanism of systematic degradation: contributors adapt to the system rather than the task, and the dataset loses precisely the complexity that makes it valuable.

The observations in this document are drawn from real project experience. Specific details have been omitted, but the patterns described are broadly recognisable and, for the most part, language-independent — the focus on Russian-language audio reflects the author's direct experience rather than the scope of the problems themselves.

The document is addressed to project managers, QA leads, and guideline authors, and examines each problem from the perspective of its root cause and practical remediation.

Note: the argument throughout this document rests on a specific assumption: that the target use case requires data that reflects natural language in use. This assumption does not hold universally — benchmark datasets, narrow classifiers, and controlled evaluation sets may legitimately prioritise consistency and clarity over naturalism. The purpose of the data determines what quality means in that context, and that distinction should inform every design decision described here.

Typical problems

1. Underspecified Instruction with an Illusion of Specification

A guideline that includes examples feels more actionable than one that does not. However, examples are not boundaries. When a rule is illustrated rather than defined, annotators and reviewers are left to infer the boundary from the examples provided — and different people draw that boundary in different places.

A common instance of this pattern in Russian audio transcription is the instruction to transcribe words "as they sound." The intent is usually to capture colloquial reduction: *зд^{ра}сь^{те}* rather than *зд^{ра}в^{ст}в^ей^{те}*, *щ^ас* rather than *с^ей^час*. But the instruction, as typically formulated, provides examples without establishing a linguistic level — colloquial norm, dialectal variation, or full phonetic transcription are not the same thing, and the examples do not distinguish between them. Without that boundary, the rule is logically extensible to spellings like *ка^{ро}ва* or *ма^{ла}ко*, which no project intends to collect. Annotators who follow the rule literally and reviewers who apply it intuitively will consistently disagree.

The same pattern appears in rules governing non-lexical sounds. A boundary dispute between *нннннн* (a lengthened onset, not a filler) and *мм, не знаю* (a filler) cannot be resolved by a rule that defines fillers by example rather than by phonetic or functional criteria. Both sounds occupy the same perceptual space; only a principled definition separates them.

Root cause: The instruction substitutes illustration for definition. Examples constrain the *centre* of a category but leave its *edges* unspecified.

Remediation: Specify the target register explicitly — name the linguistic level the instruction is operating at (e.g. informal written norm, colloquial speech) rather than illustrating it with examples alone. Where possible, provide both in-scope and out-of-scope cases to mark the boundary, not just the centre.

2. Subjective Threshold Without a Calibration Anchor

Some annotation decisions hinge not on linguistic interpretation but on perceptual judgement: is this sound loud enough to transcribe? Is this background noise significant enough to flag? Rules of this type require a threshold on a continuous scale — and a threshold, unlike a categorical definition, cannot be conveyed in words alone. Describing it as "loud," "significant," or "noticeable" does not locate it on the scale; it only restates that a scale exists. Without a shared reference point, every reviewer places the threshold independently.

A recurring example is the instruction to transcribe audible sighs but ignore quiet breathing. The rule is intuitively reasonable, but "audible" and "quiet" are not absolute values — they are relative to the listener's perception and equipment. Two reviewers applying the rule in good faith will calibrate differently, and neither can be shown to be wrong on the basis of the instruction alone.

Background noise presents the same problem with an additional technical dimension. A contributor recording in a moderately noisy environment may produce audio that registers clearly on a high-sensitivity microphone but falls below the threshold of a reviewer using standard consumer equipment. The disagreement here is not perceptual but instrumental: the same sound is literally inaudible to one party. No amount of textual clarification resolves this, because the instruction is not the source of the conflict — the absence of a shared reference point is.

Root cause: The instruction defines a threshold concept without providing a calibration anchor. Reviewers and contributors have no shared reference against which to align their judgements.

Remediation: For threshold-based criteria, reference audio examples are not optional — they are the instruction. A curated set of calibration tracks (covering in-scope and out-of-scope cases at the boundary) should be provided at project onboarding and remain accessible throughout.

3. Normative Gap: The Phenomenon Exists, the Guideline Does Not Cover It

Not all annotation conflicts arise from ambiguous rules. Some arise from the absence of a rule where one is needed. When a linguistic phenomenon is real and recurrent but falls outside the scope of the guideline, reviewers fill the gap with individual judgement — and individual judgements diverge.

A clear example from Russian audio transcription is the schwa: the reduced, centralised vowel that is pervasive in natural Russian speech but lacks a standard orthographic representation. In regular words this is not a problem — transcription follows spelling, and the phonetic reality is ignored by convention. But fillers are different: *эм*, *ээ*, *аа* are nothing but vowel sounds, sometimes followed by a nasal, and there is no orthographic form to default to. In the absence

of a guideline, some reviewers correct transcriptions with *ə* to *a*, others do the reverse — both acting in good faith, both applying different intuitions about what the "correct" representation should be.

This pattern tends to go undetected longer than other problems, because reviewers do not flag it as ambiguity — they flag it as an error. The underlying cause is invisible until the inconsistency is noticed at a higher level of QA.

Root cause: Guideline development did not account for phenomena that are linguistically real but orthographically unrepresented or normatively ambiguous in the target language.

Remediation: Guideline authoring for Russian audio should include a review of known phonetic phenomena that lack a standard orthographic representation. Where a convention cannot be established, the guideline should explicitly acknowledge the gap and specify a default — even an arbitrary one — to ensure consistency.

4. Insufficient Localisation of Instructions

Many audio annotation projects originate in English and are then adapted for other languages. The adaptation is often treated as a translation task — but translation is not localisation. A guideline that is well-specified for English audio may be underspecified, misleading, or simply inapplicable when applied to Russian, because the two languages differ not only in vocabulary but in phonology, morphology, and the very phenomena the guideline is designed to address.

The problem is compounded when the person responsible for translation is not a subject-matter expert in linguistics or audio annotation. Instructions that rely on implicit knowledge of English phonetic conventions — what counts as a filler, how reduction works, what "natural speech" means in practice — do not carry that implicit knowledge across the language boundary. The translator produces a grammatically correct Russian text that does not actually address Russian speech.

A specific failure mode: English fillers (*uh*, *um*) are themselves schwa-based sounds, and their orthographic conventions in annotation practice are relatively settled. Russian filler transcription conventions are not — and a guideline that carries over English examples without addressing the Russian inventory leaves annotators without usable guidance. The same applies to colloquial reduction examples such as *gonna*, *kinda*, *lemme*: they illustrate a real phenomenon, but one that is language-specific in its phonology, morphology, and scope. For a Russian-language task they are not just unhelpful — they imply a parallel that does not hold. The schwa problem described in the previous section is, in part, a localisation failure: the English source guideline had no reason to address it, and the adaptation did not introduce it.

Root cause: Localisation is treated as translation. Language-specific phenomena are not introduced during adaptation, and implicit assumptions from the source language are carried over unchecked.

Remediation: Localisation of annotation guidelines should involve a linguist or experienced annotator with domain knowledge in the target language. The adaptation process should include an explicit review step: which phenomena in the source guideline have no direct equivalent, and which phenomena in the target language are not covered by the source at all.

5. Conflation of Classification Axes

A well-designed classification system is built on a single principle per level: each category at a given level should answer the same question. When categories at the same level answer different questions, the system produces structural ambiguity that no amount of annotator training can resolve — because the problem is in the architecture, not in the annotators.

A straightforward example: a topic classifier that places *Chemistry* and *Programming* alongside *Advice* on the same level mixes two fundamentally different axes. *Chemistry* and *Programming* answer the question "what is this about?" — they are domain categories. *Advice* answers the question "what is this for?" — it is a functional category. These axes are orthogonal: a conversation can be about chemistry and be advisory, or about programming and be purely informational. When they share a level, every such conversation becomes unclassifiable without arbitrary disambiguation.

Root cause: Classification categories were developed by listing intuitive distinctions without first establishing the axis — the question each level is designed to answer. The result is a taxonomy that feels complete but is structurally inconsistent.

Remediation: Before assigning categories, define the axis explicitly: what question does this level answer? Categories that answer different questions belong on different axes, even if they feel thematically related.

6. Blurred Boundaries Between Adjacent Categories and the Single-Label Problem

Even a taxonomically sound classification system — one where each level answers a single, consistent question — can produce systematic disagreement if the boundaries between neighbouring categories are not explicitly defined. This is distinct from conflating axes: the categories here are on the correct axis, but their edges overlap in ways the guideline does not address.

A recurring example is the boundary between *Health/Fitness* and *Biology*. Both are domain categories; neither is misplaced. But a prompt about the mechanics of muscle fatigue, or the hormonal response to exercise, sits comfortably in either. Without a principled boundary — for instance, that *Health/Fitness* covers applied, user-oriented content while *Biology* covers explanatory or scientific content — annotators resolve the ambiguity individually, and inter-annotator agreement suffers predictably.

The problem is significantly compounded when the classification interface enforces a single label per item. A real user prompt about muscle soreness after exercise will most naturally combine both registers: "why does this happen, and what should I do about it?" — touching *Biology* in the first half and *Health/Fitness* in the second. This is not an edge case; it is the most probable form a genuine user query takes. Where multi-label assignment is possible, such a prompt can be honestly represented. Where only one label is allowed, the contributor is forced to commit to a boundary the guideline has not drawn — and the most natural, representative prompts become the hardest to classify. In practice, they are discarded, producing a dataset skewed toward artificially simple, single-topic prompts that are easy to label but poorly representative of real user behaviour.

Root cause: Category definitions describe the centre of each category but leave the edges unspecified. In single-label systems, this forces arbitrary decisions on borderline cases and systematically excludes multi-topic content.

Remediation: For each pair of adjacent categories where overlap is foreseeable, the guideline should provide a distinguishing criterion — a decision rule for cases that could plausibly belong to either, not just examples of each category in isolation. Where the task permits, multi-label assignment should be considered as a structural solution rather than a workaround.

Systemic Consequences: How the Problems Compound

The problems described in the preceding sections do not operate independently. They interact through a self-reinforcing mechanism that degrades dataset quality in a direction that is difficult to detect and difficult to reverse.

When instructions are underspecified and thresholds are uncalibrated, reviewers face genuine ambiguity. In the absence of shared criteria, the path of least resistance is to flag anything non-standard as an error. Contributors learn this quickly. A prompt or recording that is complex, multi-layered, or linguistically rich carries a higher risk of rejection than a simple one — not because it is wrong, but because it is harder to evaluate. Over time, contributors optimise for reviewability rather than quality: prompts become simpler, speech becomes more careful, natural features get removed. The dataset becomes internally consistent and easy to process, while systematically losing the properties that make it valuable.

This dynamic has a compounding effect on reviewer quality. When simple, unambiguous submissions consistently receive high scores, the contributors who produce them advance. In many projects, reviewers are drawn from high-performing contributors — but high performance in a degraded system selects for people who are good at avoiding complexity, not at recognising it. The result is a reviewer pool that is structurally ill-equipped to evaluate nuanced content, and whose feedback further reinforces the simplification pressure on contributors.

The feedback mechanisms that might correct this tend to be formally present but practically ineffective. Contributor appeals exist but rarely change outcomes. Systematic patterns in reviewer disagreement are not aggregated or acted upon. The system is optimised for throughput, and quality signals that require interpretation are absorbed without response.

The clearest evidence of this dynamic is what disappears from the dataset. In audio projects, natural speech features — fillers, reduced forms, hesitations — are edited out because they generate reviewer disputes. In prompt collection, multi-topic queries are discarded because the classification interface cannot accommodate them. In both cases, the dataset loses precisely the content that reflects how people actually speak and write — which is the content the model most needs to handle.

Recommendations

1. Writing and Localising Guidelines

The most common failure in guideline authoring is substituting illustration for definition. Examples are useful for orienting annotators, but they describe the centre of a category, not its edges. Every guideline that governs a gradient or variable phenomenon — reduced speech forms, non-lexical sounds, background noise — should include an explicit boundary condition: a statement of what falls outside the rule, not just what falls inside it.

For rules governing speech reduction and colloquial forms, the boundary condition should take the form of a named register. Specifying that transcription follows informal written norm — the way a native speaker would render the word in a casual message — is more constraining than any list of examples, because it gives annotators a reference system they already know how to apply. Examples can then illustrate the register rather than attempt to define it exhaustively. Instructions phrased as "transcribe as pronounced" or "write as it sounds" should be avoided entirely: they imply a phonetic standard without specifying one, and are logically extensible to levels of reduction no project intends to collect.

Where textual description is structurally insufficient — as it is for any threshold-based criterion — audio reference examples are not optional. A curated set of calibration tracks, covering both in-scope and borderline cases, should be prepared before the project launches and remain accessible throughout. This applies to audible sighs, background noise levels, and any other criterion where the decision hinges on a perceptual judgement rather than a categorical one.

Localisation of guidelines into another language should never be treated as a translation task. The adaptation should be led or reviewed by a linguist or experienced annotator with domain knowledge in the target language, and should include an explicit gap analysis: which phenomena in the source guideline have no direct equivalent, and which phenomena in the target language are not covered by the source at all. Examples drawn from the source language — colloquial reduction forms, filler conventions — should be replaced with target-language equivalents, not translated literally.

2. Designing Classification Systems

Before assigning any categories, define the axis: the single question that every category at a given level is designed to answer. Categories that answer different questions — domain versus function, topic versus style, subject matter versus intended use — belong on different axes and should never share a level. This decision should be made explicitly, in writing, before category names are assigned, because mismatches that are obvious in the abstract become invisible once a list of category names is in front of you.

For adjacent categories where boundary overlap is foreseeable, a distinguishing criterion should be provided alongside the category definitions — not examples of each category in isolation, but a decision rule for cases that could plausibly belong to either. If no clean boundary exists, that is a signal that the categories may need to be restructured rather than better defined.

The question of single-label versus multi-label assignment should be resolved at the design stage, not treated as a default. Single-label systems are operationally simpler, but they impose a hidden cost: they force arbitrary decisions on borderline cases and systematically exclude prompts that reflect real user behaviour. If the task requires single-label assignment for downstream reasons, this constraint should be explicit, and contributors should be given clear guidance on how to handle multi-topic content — including whether such prompts should be simplified, split, or discarded, and why.

3. Setting Up QA and Reviewer Systems

Reviewer calibration is not a one-time event at project onboarding. It is an ongoing process that requires shared reference material, regular alignment sessions, and a mechanism for surfacing systematic disagreement. Otherwise, standards drift — ambiguous cases get resolved individually, and individual resolutions quietly diverge over time.

Calibration material should mirror the calibration material provided to contributors: the same audio reference tracks, the same boundary cases, the same decision rules. A reviewer who has not heard the reference tracks cannot reliably evaluate a contributor's threshold judgements. Where new edge cases emerge during the project, they should be documented, discussed, and added to the shared reference set — not resolved ad hoc and forgotten.

Reviewer selections should not be based solely on contributor performance scores. High scores in a system with underspecified guidelines may reflect an ability to avoid complexity rather than an ability to recognise quality. Reviewer candidates should be assessed specifically on their handling of ambiguous and boundary cases — the cases that matter most for inter-annotator agreement.

When a reviewer decision is formally found to be incorrect, the correction must be applied retroactively to the contributor's record. A system that acknowledges a reviewer error without remedying its consequences sends an unambiguous signal to contributors: the process is not fair, and appealing decisions is not worth the effort. The downstream effects — disengagement, reduced care, accelerated simplification — are predictable and significant. Retroactive correction is operationally complex, but its absence is not a neutral choice.

4. Communicating the Task to Contributors

Contributors who understand the purpose of their work make better judgements on edge cases — but only if the evaluation system supports rather than contradicts that understanding. Explaining that natural speech features are valuable while penalising them through QA sends a conflicting signal, and the economic signal will always win. Task communication is therefore not a standalone measure; it is effective only in conjunction with a calibrated, well-specified QA process.

Within those conditions, the necessary level of transparency is modest: not what model is being trained, or for whom, but what kind of data is valuable and what kind of simplification undermines it. This gives contributors a principle to reason from when the rules run out — which, in any natural language task, they inevitably will.

5. Cost and Timing

The problems described in this document are significantly cheaper to prevent than to correct. Underspecified guidelines, missing calibration material, and poorly designed classification systems produce costs that are invisible at the point where they are incurred — a disputed annotation, a discarded prompt, an unresolved edge case — but accumulate into outcomes that are expensive to address: systematic reviewer drift, dataset bias, and in the worst case, a corpus that needs to be partially or wholly recollected.

The upfront investments that prevent these outcomes are modest in comparison. A careful gap analysis during guideline authoring, a set of audio calibration tracks prepared before launch, a classification design review before categories are finalised — none of these are large undertakings relative to the cost of a full project. They require time from people with the right expertise, which means they require planning. The constraint is not budget but sequencing: these decisions cannot be made well under time pressure, and they cannot be revisited cheaply once a project is in motion.

Feedback from contributors and reviewers tends to arrive when it is already operationally difficult to act on. This is partly structural — problems become visible through volume, and volume takes time to accumulate — and partly cultural: feedback channels exist but are not always designed to produce action. The more useful investment is in mechanisms that surface problems early: inter-annotator agreement tracking from the first batches, regular reviewer alignment sessions, and a process for escalating edge cases that do not fit existing guidelines rather than resolving them ad hoc.

Conclusions

The problems documented here share a common origin: QA processes designed for operational efficiency without sufficient consideration of linguistic reality. Efficiency is a legitimate goal — annotation projects operate under real constraints of time and budget — but a system optimised purely for throughput and reviewability will systematically produce the wrong kind of efficiency. It will process data quickly and consistently while quietly removing the properties that make the data valuable.

The mechanism is not malicious and not obvious. No individual decision in the chain — flagging an ambiguous transcription, discarding a multi-topic prompt, selecting a high-scoring contributor as a reviewer — is unreasonable in isolation. The damage is done by the accumulation of individually defensible decisions within a system that was not designed to handle the full complexity of natural language.

The practical implication is that quality in annotation projects cannot be fully delegated to QA. A QA system can enforce consistency; it cannot substitute for well-designed guidelines, sound taxonomy, and calibrated reviewers. These require investment at the design stage — before the project launches, before the guidelines are finalised, before the first batches are reviewed. The cost of that investment is visible and immediate. The cost of skipping it is distributed, delayed, and substantially higher.